Socio-economic Detriments for Maternal Mortality: A Cross-demographical analysis

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Introduction:

Maternal Mortality Rate comprises of loss of life during and after pregnancy. There are many contributing factors, including but not limited to, health and socio-economic conditions. According to WHO, the unacceptable high rate, in most cases could have been prevented[1]. Contributing factors include economic conditions, health expenditure and overlooked health conditions, due to unavailability of better living circumstances and/or lack of education. The objective of this study is to analyse socio-economic factors and health resources across countries for the year 2019 and to quantify the relationship between the selected independent variables against Maternal Mortality Rate using regression analysis.

Data Description:

The dependent Variable, Maternal Mortality rate is transformed as log. The data for analysis was retrieved from World Bank Open Data API for the year 2019 using the wbdata python interface. The independent variables considered are as follows

Variable	Description	Expected Coefficient Sign
log_gdp	GDP per Capita	negative
Current_Health_Expenditure	Health expenditure as a percentage of GDP	negative
Anemia	Prevalence of anemia among women of reproductive age (15 to 49)	negative
Adult_Literacy_Rate	Percentage of literate adults	negative

After cleaning null values using wbdata.dropna(), the final dataset consists of 94 observations.

Transformation:

Applied Log transformation to both Maternal Mortality Rate (log_MMR) and GDP per Capita (log_gdp) as both these values are heavily skewed. This helps in reducing skewness and stabilizing variance.

The Statistical Model used is ***Multiple Liner Regression*** employing ***OLS method***.

$\log(MMR_i) = b_0 + b_1 \log(GDP_i) + b_2 \text{ HealthExp} + b_3 \text{ Anemia} + b_4 \text{ Adult_literacy_rate}$

Variables Evaluation:

Variance Inflation Factors(VIF) was calculated for the independent variables against the dependent variable - Maternal Mortality Rate, to evaluate the multicollinearity between them. The results are as follows

Variable	VIF value
const	198.690835
log_gdp	2.543047
Current_Health_Expenditure	1.217635
Adult_Literacy_Rate	3.112754
Anemia	2.423751

All the variables exhibited a VIF value under 5, this ensures that significant multicollinearity doesn't exist.

Regression Result and Interpretation:

The OLS regression results are as follows

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                            OLS Regression Results
==============================================================================
Dep. Variable:                log_MMR   R-squared:                       0.669
Model:                            OLS   Adj. R-squared:                  0.654
Method:                 Least Squares   F-statistic:                     45.00
Date:                Mon, 30 Mar 2026   Prob (F-statistic):           1.30e-20
Time:                        17:07:39   Log-Likelihood:                -112.47
No. Observations:                  94   AIC:                             234.9
Df Residuals:                      89   BIC:                             247.6
Df Model:                           4
Covariance Type:            nonrobust
==============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
const                         11.9529      1.196      9.993      0.000       9.576      14.330
log_gdp                       -0.6252      0.128     -4.884      0.000      -0.880      -0.371
Current_Health_Expenditure     0.0515      0.032      1.611      0.111      -0.012       0.115
Anemia                         0.0050      0.013      0.390      0.698      -0.020       0.030
Adult_Literacy_Rate           -0.0327      0.009     -3.748      0.000      -0.050      -0.015
==============================================================================
Omnibus:                       23.825   Durbin-Watson:                   1.700
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               43.708
Skew:                          -0.999   Prob(JB):                     3.23e-10
Kurtosis:                       5.677   Cond. No.                     1.27e+03
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As the Adjusted R^2 value is 0.654, about 66% of Obstetric Mortality contributing factors for the year 2019 have been explained using this model. The p-value is reported as 0.000 for GDP per Capita and Adult Literacy Rate factors, suggesting statistical significance. Whereas Health Expenditure and Anemia have p-values greater than 0.05. The values are

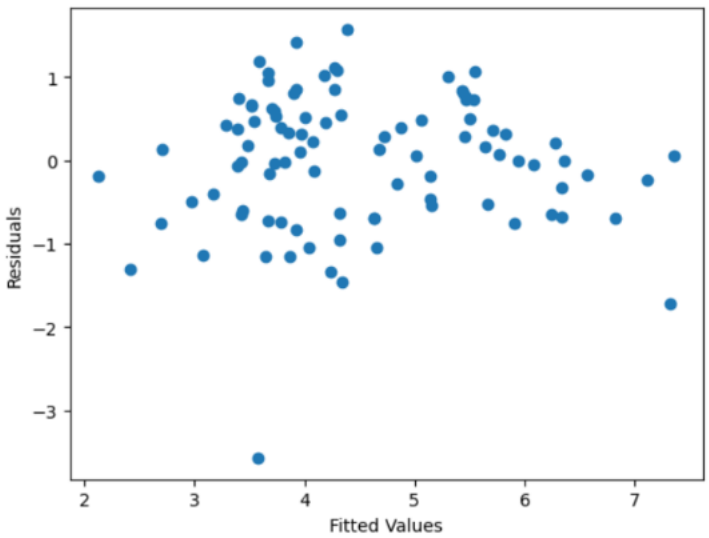
0.111 and 0.698 respectively, this provides evidence against the null hypothesis, leading us to conclude that these variables do not have a significant relationship with the dependent variable under consideration.

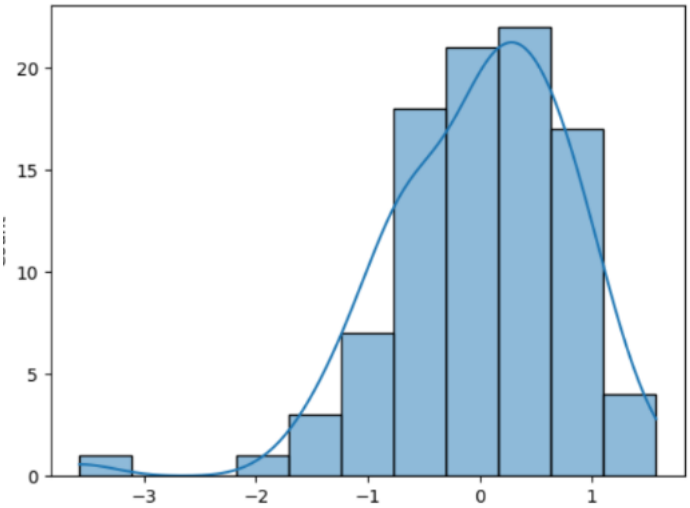
The coefficients for GDP per capita and adult literacy rate have an inverse relationship with the dependent variable. Therefore, for every 1% increase in GDP per capita, pregnancy-related mortality decreases by 0.625% and for every 1 percentage point increase in Adult Literacy Rate, pregnancy-related mortality decreases by 0.0327%. These expected results are assumed that all other variables are held constant. The other two independent variables do not have statistically significant associations; this may suggest that socioeconomic conditions prevail when applying cross-demographic analysis.

Though significant results are provided by R^2 value and the coefficients, the Jarque-Bera value is significantly high. This indicates that there are non-normal residuals which may suggest the presence of outliers. However, considering that the sample size is small (94) and the analysis is done on cross-country data, OLS estimates hold.

Residual Diagnostics:

Residual analysis was conducted using Residuals Vs Fitted Values Plot and Histogram. The plots are constructed using matplotlib and seaborn interfaces respectively.





In the Residuals Vs Fitted Values, the residuals are scattered randomly around zero, this reasonably satisfies the linearity assumption. The scatter plot further suggests that there are some outliers present[2]. In the residual histogram, the residuals are roughly clustered around zero and are approximately bell shaped in the centre. At the negative value, we can see a longer tail, indicating the presence of extreme outliers. As the skew value is approximately equal to 1, it suggests that some countries are performing far better than the proposed model expects them to. The results suggest leptokurtic with Kurtosis value at 5.677 which leads us to deduce that there are extreme residuals than the normal distribution expects. These findings infers that the considered factors do not have the predicted inverse linear relationship for all countries.

Conclusion:

Based on the results, GDP and Adult Literacy Rate have an active role in reducing the Maternal Mortality Rate, and health expenditure per capita and adult literacy rate do not have an impactful control over Maternal Mortality when considering cross-demographical data. The outliers indicate that some countries stray away from the predicted pattern in this model and that there are more contributing factors for Maternal Mortality Rate. These suggest that in some cases the rates are much worse or far better than expected. A further detailed analysis and understanding of the detriments can be arrived at by analysing changes over the years in a specific country or demographical region.

References:

1 https://www.who.int/news-room/fact-sheets/detail/maternal-mortality , World Health Organization.

2 https://online.stat.psu.edu/stat462/node/117/ , Penn State Eberly College of Science, Applied Regression Analysis.